Improving Vision-Language Models through Evolutionary Chain-of-Thought Prompting

John Rho, Regina Lin

Department of Applied Mathematics, Harvard University

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INTRODUCTION

Vision-Language Models (VLMs) excel in multimodal tasks but face challenges in visually ambiguous scenarios, leading to reasoning inconsistencies and errors. Chain-of-Thought (CoT) prompting has improved reasoning in language models, but its application to VLMs remains limited. Static CoT prompts often fail in tasks requiring nuanced, context-specific reasoning. This study introduces Evolutionary Chain-of-Thought (EoT) prompting, a dynamic approach that tailors prompts for improved reasoning accuracy. We evaluate EoT on the Japanese Uncertain Scenes (JUS) dataset, focusing on tasks such as object counting and image recognition.

Key Contributions

- Developed an Evolutionary CoT framework for vision-language tasks.
- Improved reasoning on the JUS dataset, known for its visually ambiguous scenarios.
- Introduced crossover and sophisticated dual-mutation operations for optimizing prompts for complex image recognition tasks
- Demonstrated superior accuracy and reduced variance compared to static CoT and baseline methods.

BACKGROUND

Model Selection

We use GPT-4o, OpenAI's multimodal model, designed to process text, voice, and vision efficiently with a 128,000-token context window. For cost efficiency, GPT-4o Mini is employed for text-only tasks.

Zero-Shot Chain of Thought Prompting

Chain-of-Thought (CoT) prompting guides models to decompose problems into step-by-step reasoning, improving intermediate logic and decision-making. Inspired by previous methods like Kojima et al. (2022), we refine the original question Q by appending a structured CoT instruction TT, such as:

Original Prompt:
"How many food items are shown in this photo?"
CoT Prompt:
"Please provide a detailed step-by-step reasoning for your answer. Ensure that your reasoning is clear and follows a logical progression."

Evolutionary Chain of Thought (EoT) Prompting

Static CoT prompts may degrade performance in complex tasks. To address this, the Evolutionary Chain-of-Thought (EoT) prompting framework, introduced by Jin, Liu, and Tan (2024), applies evolutionary algorithms to dynamically optimize prompts for large language models (LLMs).

- **Crossover & Mutation:** Parent prompts are combined and modified to generate diverse and effective prompts.
- **LLM-Driven Selection:** The "most suitable" prompt is chosen to rewrite the original problem, ensuring improved reasoning.

METHODS

Japanese Uncertain Scenes (JUS) Dataset

- 39 visually complex images from Japan.
- Tasks include object counting, food recognition, and cultural site identification.
- Labeled with ground truth answers for evaluation.

Base Prompt Initialization

- Two base prompt types: 'Counting' and 'Descriptive'.
- Additional variations to increase diversity, e.g., "Be specific and detailed."

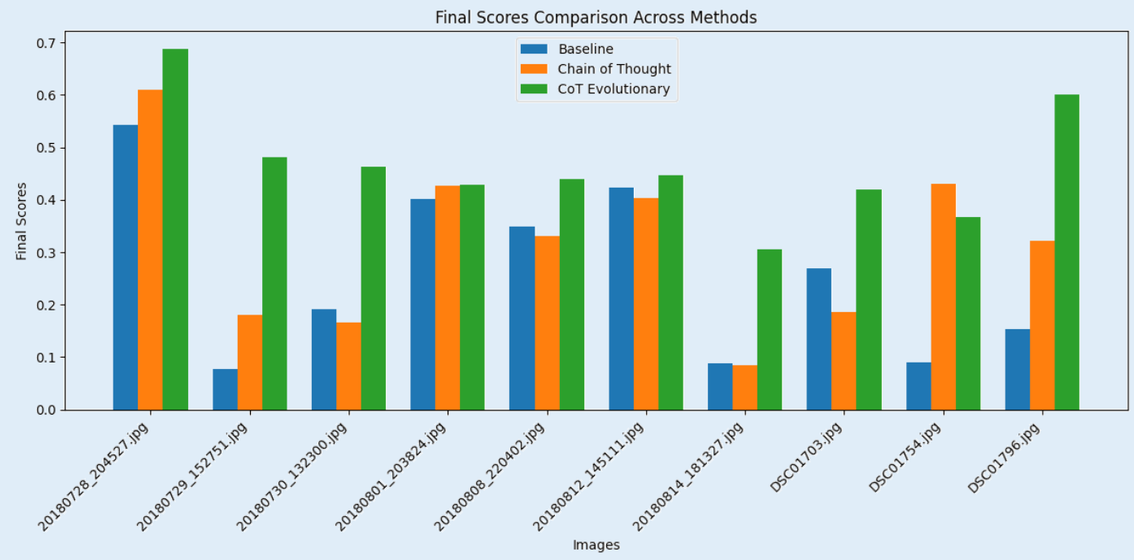
Elite Preservation

- Top 20% of prompts retained each generation.
- Parents selected from elites for crossover and mutation:
  - Crossover: Combines instructions and steps from parent prompts.
  - Mutation: Enhances prompts through targeted refinements.

Dual-Mutation System

- Uses GPT-4o Mini to analyze prompt performance and identify issues.
- Tailored enhancements based on four common issue patterns:
  - Counting: Systematic grids, sequential counting.
  - Verification: Double-checking and cross-validation.
  - Structural: Clear phases (assessment, counting, verification).
  - Uncertainty: Explicitly address visibility issues and confidence levels.

RESULTS



Performance Comparison:

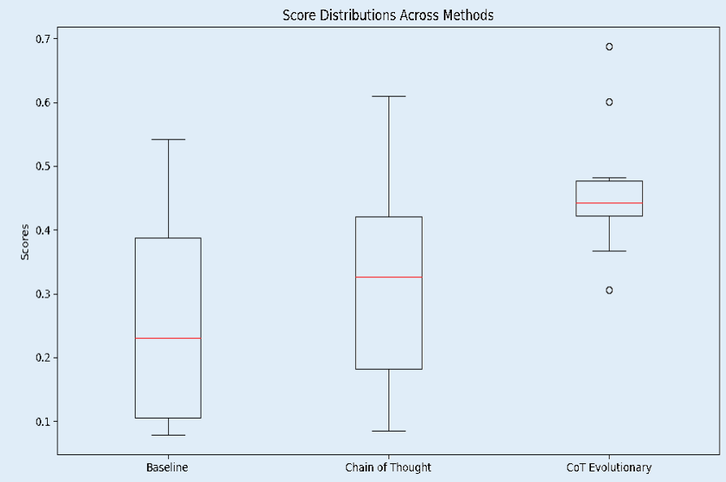
- Baseline direct VLM queries vs. Zero-Shot CoT vs. Proposed Evolutionary CoT methods
- EoT significantly outperforms baseline and Zero-Shot CoT methods in 9/10 tasks.
- Average Score Improvement:
  - +0.2 points over baseline.
  - +0.1 points over Zero-Shot CoT.

Failure Analysis:

- Outlier case "DSC01754.jpg" revealed limitations in analyzing highly obscured objects, where static CoT outperformed EoT.
- Highlights the need for additional robustness checks in the evolutionary framework.

Variance Reduction:

- EoT achieves reduced variance, indicating more consistent performance across tasks.
- The guided mutation process aligns prompts more effectively with task-specific requirements.



DISCUSSION

The results demonstrate that Evolutionary CoT prompting significantly enhances reasoning capabilities in VLMs. By dynamically optimizing prompts, the EoT framework provides:

- Post-training improvements without expensive retraining or fine-tuning.
- Alignment of VLM outputs with real-world expectations using labeled ground truth.
- Practical applications for domain-specific tasks such as medical imaging or autonomous driving.

The intelligent mutation system ensures prompts evolve iteratively, improving reasoning accuracy and reducing variance. These findings highlight the potential of adaptive prompting as a cost-effective alternative to traditional retraining.

Limitations

- Computational intensity of evolutionary algorithms.
- Experiments limited to a fixed population size and one VLM model (GPT-4o Mini).

Future Directions

- Experimenting with larger datasets and diverse VLM architectures.
- Exploring initialization strategies for improved evolutionary trajectories.

REFERENCES

Groot, Tobias, and Matias Valdenegro-Toro. Overconfidence Is Key: Verbalized Uncertainty Evaluation in Large Language and Vision-Language Models. Department of Artificial Intelligence, University of Groningen.

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